

Evidence-Based Training Recommendations for the Elite Judoka

Dale M Harris ¹, Steven Foulds ², Christopher Latella ³

¹Inspire Institute of Sport (IIS), Vijaynagar, Bellary, India

²School of Exercise and Nutrition Sciences, Deakin University, Geelong, Australia

³Centre for Exercise and Sports Science Research (CESSR), School of Medical and Health Sciences, Edith Cowan University, Joondalup, Australia

Corresponding Author: Dale M Harris

Address: Inspire Institute of Sport, Vijaynagar, Bellary, India 583275

Ph: +91 94806 94069

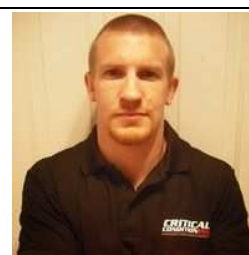
Email: dale.harris@inspireinstituteofsport.com

Word Count: 4,169

Mr. Dale Harris is the Head of Strength and Conditioning at the Inspire Institute of Sport, India.



Mr. Steven Foulds is an honours student and currently teaches coaching at Deakin University, Australia.



Dr. Chris Latella is a post-doctoral researcher in the School of Medical and Health Sciences at Edith Cowan University.



Conflicts of Interest and Source of Funding: The authors report no conflicts of interest and no source of commercial funding and no funding outside of Australia or India.

Abstract

There is a paucity of sport-specific evidence regarding strength and conditioning strategies to improve elite Judo athletes (EJA) performance. Therefore, the aim of this article is to evaluate the current literature and present evidence-based training recommendations for EJA. Based on the sport-specific demands, appropriately sequenced periodized strength and conditioning programs that include; 1) heavy strength training (HST), 2) power training (i.e., contrast loading [CL], cluster sets, weightlifting [WL] and plyometric exercises), and 3) high intensity interval training (HIIT), may provide sufficient stimuli to improve upper and lower-body strength, aerobic and anaerobic power for EJA.

Key words

Strength; power; martial arts; periodization; conditioning; high-intensity; combat

Introduction

Judo is a Japanese martial art conceptualized in the 19th century and first introduced as a combat sport at the Tokyo 1964 Olympic Games. The high-intensity, intermittent nature of Judo requires speed, power and strength to perform a multitude of ground and stand-up defensive and attacking grappling techniques. Elite Judo athletes (EJA) often compete in multiple matches during a single day of competition, with each match capped at five minutes. However, matches may be concluded early if the athlete obtains a full point (*ippon*), which can be achieved via three methods; 1) the opponent is flipped onto their back with force, 2) the opponent is pinned for more than 20 seconds (s) on their back (*osaekomi-waza*), or 3) the opponent is forced into submission (*shime-waza* or *kansetsu-waza*). If scores are tied, an unlimited 'golden point' period ensues, whereby the first athlete to score a point is declared the winner. A single match may last from a few seconds to well beyond 10 minutes, however, the mean length is approximately three minutes (16). Therefore, strength and conditioning professionals should carefully consider the specific physiological and performance demands required by EJA to facilitate successful performance.

Recently, research has attempted to investigate the numerous physiological characteristics underpinning success in judo during training and/or competition (16, 18, 20). Specifically, anaerobic and aerobic capacity are thought to play a key role in judo success (34, 38). Additionally, upper and lower-body strength and power are also considered to be important neuromuscular traits to facilitate successful judo performance (16, 18, 20). However, although judo-specific physiology research has increased in the past decade, strength and conditioning training recommendations remain scarce.

To our knowledge only a handful of evidence is available regarding Judo-specific strength and conditioning practice (1, 17). To further highlight the issue, the available studies have either provided recommendations based on general periodization literature (1), or provided strength and conditioning strategies that were not entirely judo specific. For example, Franchini et al (17) prescribed continuous running as part of a judo conditioning program, which fails to address some key aspects of anaerobic performance specific to EJA (19, 20). Therefore, although these initial studies provide some insight into strength and conditioning practice for EJA, evidence based, sport-specific strategies and recommendations are desperately required for athletes and professionals alike.

Therefore, the aim of this article is to; 1) summarize the current literature investigating the bioenergetic and neuromuscular evidence underpinning EJA performance, and 2) provide updated evidence-based training recommendations and strategies to facilitate the physical development of EJA. The findings are expected to be particularly important for strength and conditioning professionals implementing general and specific preparatory programs for EJA across a competitive season.

Warm-Up

The current knowledge surrounding warm-ups and injury prevention methods for combat sports is poor. For judokas, the commonly injured areas are the elbow, knee and shoulder (36). Therefore, these joint structures should be considered when developing warm-up routines. Specifically, these warm-up routines should consist of both global and joint-specific range of motion (ROM) activities to prepare for both resistance training movement patterns and mat-based training. Based on previous wrestling research (26) and more generalized warm-up recommendations (23), a warm-up lasting approximately 10-15 minutes is considered optimal. In addition, the '9+ program', developed by Maliaropoulos et al. (31), consisting of nine individual and partner-based judo-specific warm-up exercises should be considered as part of a judo-specific warm-up. Although the efficacy of the program for attenuating injury prevalence in judo remains elusive, this program can be adopted in a safe and time-effective manner.

Bioenergetic Requirements

Due to the widespread technical and physiological requirements of judo, all three metabolic energy systems (i.e., aerobic, anaerobic glycolytic [lactic] and ATP-CP [alactic]) are utilized during a match to provide energy and aid recovery between high intensity efforts (20). Of the limited evidence available in this area, Franchini et al. (16) investigated the energy expenditure of three stand-up judo techniques using 10-30 s all-out work periods that mimicked judo competition (i.e., *Uchi Komi*). Blood lactate, post-exercise oxygen consumption and the total absolute energy (kilojoule) expenditure did not differ between each grappling technique. However, no comparison was made to ground-based grappling techniques, despite competitive matches commonly progressing to the ground (16, 18). Therefore, given the importance of standing and ground-based techniques in judo it is prudent to understand the bioenergetic requirements of both. To our knowledge, only one study has compared standing to ground-based actions (43). Overall, higher blood lactate was reported

in the standing group (11.3 mmol.l⁻¹) compared to the ground-based group (7.7 mmol.l⁻¹), which suggests lower glycolytic demands for grappling actions performed on the ground.

A key factor in energy system contribution is the work-to-rest ratio (W:R). Franchini et al. (16) established that the W:R ratio for judokas ranges from 2-3:1 during competition with each grappling technique generally lasting 20-30 s alternated with 10 s recovery intervals between high-intensity efforts. Evidence from other combat sports suggests that during repeat high-intensity efforts there is a down-regulation of the lactic system and a greater contribution from the aerobic energy system (9, 11). However, in contrast, during the special judo fitness test (SJFT); a measure of energy system contributions for experienced male judokas, Franchini et al. (20) found that the alactic system was the predominant bioenergetic supplier. Considering these findings, the larger alactic contribution may be a differentiating factor between judo athletes and other combat sports athletes, warranting the need for further judo specific research and consideration when designing appropriate conditioning programs.

Strength Training

It is well established that heavy-strength training (HST), performed above 80% of one repetition maximum (1-RM) can improve maximal strength in elite athletes (12, 23, 46). HST has been shown to recruit high threshold motor units, facilitate musculotendinous stiffness and increase motor unit firing frequency (29). Implementing HST prior to the development of muscular power provides a sequential approach when periodising programs for both junior and senior athletes across multiple sports within a competitive season (23). Specifically, the implementation of general and task-specific HST before progressing to higher velocity training cycles (i.e., plyometrics and weightlifting) in the later stages of periodized programs is imperative to facilitate an improvement in the rate of force development (RFD) (29).

Periodization literature in other combat sports has also recommended HST to improve absolute/relative strength and RFD (10, 28) however, limited evidence is available for EJA. One such study (17), investigated the effect of 18 weeks of structured HST in EJA and found significant improvements in upper body strength, but no change in lower body power. Interestingly, lower body strength was not measured, despite many judo grappling techniques requiring the summation of force from the lower body and core into the upper torso for the “throw”. Additionally, the program was divided into two phases (general phase for 7 weeks and special phase for 11 weeks, respectively), with pre-determined loading paradigms and no

reported within phase progression. Therefore, it is unclear if systematic progression may have facilitated greater strength development across the intervention. As such, the daily adjusted progressive resistance exercise technique can assist in the acute management of training load and fluctuations in strength performance (52), and thus should be considered to more accurately prescribe appropriate individualized daily training volumes and intensities for EJA.

Due to the repetitive throwing techniques required, trunk strength is imperative for EJA. In support, Barbado et al. (5) demonstrated greater trunk extensor strength and stability in medallists compared to non-medallist EJA. Additionally, many judo grappling techniques require high-velocity pulling strength, with concurrent multi-planar trunk rotation actions performed at various angles. Helm et al. (25) found that both judo-specific ergometer pulling strength training and partner-based pulling exercises were effective at improving strength and power. Therefore, these results provide preliminary support for resistance training specificity to induce effective task-related strength transference to technical judo movements.

Recent literature has also proposed that strength adaptations are joint-angle specific (38). That is, if partial range of motion (ROM) is trained (i.e., quarter or half-squat), tendon stiffness and motor unit recruitment patterns develop for that particular ROM. Given that judo athletes perform grappling techniques in both semi and full hip flexion and extension throughout a given bout, it is strongly recommended that both partial and full ROM lower-body compound strength tasks are included to facilitate global and task-specific strength adaptations.

Additionally, it is now recognized that eccentric muscle strength is important to optimize power for EJA; particularly in the ability to repetitiously produce peak lower body power during throwing techniques (18). Furthermore, eccentric exercise may potentially reduce the prevalence and severity of non-contact injuries during repeated explosive muscle actions (41). Although typical judo injuries are joint specific torsional events (i.e., twisting of the knee and ankle) (36), eccentric strength training should still be considered in conjunction with any lower body strength and power development program to minimize non-contact soft tissue injuries.

A holistic consideration for strength and conditioning professionals is the overall strength of the athlete. Suchomel et al. (46) has suggested that athletes should aim to achieve a strength

reserve, whereby the resistance able to be lifted is approximately ≥ 2.0 times their body mass. It must be noted though that this value is based on the back squat and there is limited evidence to support whether the same level of strength is needed in other compound lifts. To our knowledge, no research to date has investigated different individual relative strength levels on performance outcomes in EJA. Nevertheless, it is anticipated that greater relative strength will contribute to improved power development and potentially translate to sports-specific judo performance.

Power Training

The development of power is largely attributable to various neural (i.e., motor unit recruitment, firing frequency and synchronisation) and morphological (i.e., changes in cross-sectional area, fascicle length and pennation angle) adaptations (12). Training at different zones along the force-velocity curve is now a widely accepted approach to improving peak power output (PPO) (12, 45). However, there is currently no consensus regarding the optimal load to maximize PPO adaptations, especially for EJA. In particular, the discrepancies are likely due to the differences in employed methodology within research study designs (21, 34). None the less, it appears that training at both low and high percentages of 1RM produce superior PPO adaptations for stronger, compared to weaker athletes. Stone et al. (44) reported that stronger athletes produce PPO at greater percentages of 1RM (~40%) in the jump squat, compared with weaker athletes (~10%). Despite the lack of uniformity, it is suggested that PPO can be produced when working against external loads that equate to 40% to 70% of 1RM during upper body exercises (2, 4) and 40% to 80% of 1RM for lower body exercises for elite athletes (2, 4). Therefore, corresponding loads should be considered in programs aimed at optimizing PPO in junior EJA due to differences in strength capabilities compared to other sports, and younger training age than previous studies (4,7). It is also important to note that PPO may be developed at higher intensities for senior EJA, with more research required at this stage.

Plyometrics

Rapid stretch-shorten cycle muscle actions are highly important for EJA performance, particularly in the execution of throwing techniques (34), and may be improved through plyometric training. However, due to the heterogeneity of plyometric interventions (37), it is difficult to ascertain a dose-response relationship for plyometric training across a variety of groups including female, youth, individual and combat sports athletes. In support of the

efficacy of plyometrics, Fukuda et al. (21) demonstrated increases in power (26.7%), force (7.7%), and velocity (19.0%) during the countermovement jump (CMJ) following 4-weeks of integrated plyometric training in adolescent EJA. Unfortunately, the type of plyometric exercises used were not disclosed, nor were prescription guidelines presented. Additionally, earlier research by Takahashi et al. (48) presented plyometric prescription guidelines for judokas, however the recommended volumes appear substantially higher than that reported in modern plyometric literature (14) and thus, should be interpreted with caution. Despite the apparent need for further evidence in EJA, based on the efficacy of plyometrics in other sports (37) it is recommended that plyometrics are integrated within a power-development phase using established guidelines (See Davies et al (14)).

Contrast Loading

Contrast loading (CL) combines high- and low-load efforts within a single set, whereby the objective is to lift a heavy resistance (i.e., >80% of 1RM) immediately followed by an open chain, high velocity movement (i.e., plyometric activity). In particular, the potentiation effect following a heavy resistance set can improve PPO during the proceeding high velocity activity if performed correctly. For example, Hammami et al. (24) used a common pairing of heavy squats followed by CMJ to examine the effects of CL, compared to HST and plyometric training alone. CL produced equivalent and/or significantly greater power outcomes across sprint, agility, repeated change of direction and jump measures over an eight-week training period in football athletes. Further, Campos et al. (8) established that isometric loading (squats) can also potentiate the performance of EJA in the CMJ and SJFT. Given the nature of throwing techniques such as *ippon seoi nage*, whereby there is a combination of high velocity hip and knee extension required, it is recommended that both HST and isometric squats form part of CL routines to develop power during a specific preparatory phase(s) of a competitive season for EJA.

Cluster Sets

Recently, there has been growing interest in the use of intra-set rest periods (i.e., cluster sets) to maintain power output during a resistance training session. Cluster sets can also be used to add overload while minimizing fatigue during specific mesocycles within a competitive season (49). It has recently been shown that intra-set rest period paradigms using a rest interval of 15-30 s can maintain velocity and power output across high volume training sessions (49). For EJA, cluster sets can attenuate cumulative fatigue, and improve lifting

quality while maintaining higher absolute training loads (27). It is suggested that cluster sets be used during power development phases throughout a competitive season for EJA.

Weightlifting

Weightlifting movements and derivatives are considered an efficacious tool to improve neuromuscular power across many sporting disciplines (45, 47). However, although weightlifting movements are biomechanically comparable to sport-related performance measures (i.e., vertical jump), Franchini et al. (17) found no benefit of 11-weeks of HST and combined weightlifting on CMJ performance for EJA. Although, upon further inspection, high velocity weightlifting tasks using lighter training loads to develop speed-strength were not prescribed within this program. As such, although loads of 90% 1RM may have corresponded to an improvement in strength, lighter loads may have elicited greater adaptations in RFD and movement speed which are likely more applicable to the actions performed during judo competitions (12, 45, 47).

Anaerobic and Aerobic Conditioning

Aerobic power values have previously been presented for male (between 44.5 – 56 ml.kg.min⁻¹) and for female (40 – 45 ml.kg.min⁻¹) EJA (18). Although these values are typically lower than aerobically dominant sports, as aforementioned, EJA rely heavily on concurrent anaerobic energy contributions. In particular, training at supramaximal intensities (i.e., anaerobic speed reserve) is a potent training stimulus that can facilitate improvements in anaerobic work capacity for intermittent high intensity sport athletes (7). To establish anaerobic speed reserve, a maximal aerobic speed test must first be performed (3). The total distance and times are recorded and converted into m/s (known as $vV02max^{-1}$), which is then used to program for maximal and supramaximal intensities (3, 7), generally via high intensity interval training (HIIT).

HIIT is an effective means of improving both anaerobic and aerobic capacity via bouts of intermittent supramaximal work intensities, without compromising concurrent strength or hypertrophy adaptations (7). For EJA, HIIT provides a time-effective training modality both within strength and conditioning sessions, and as part of *Uchi Komi* (19). Furthermore, it is common for EJA to compete at regular intervals throughout a calendar year, and therefore, HIIT can be employed either between subsequent competitions and/or during the taper phase before major competitions. Improvements in upper body aerobic power, upper and lower-

body anaerobic peak power and running velocity have also been demonstrated after 4 weeks (36) and 6 weeks (6) of HIIT in EJA. Collectively, these studies suggest that short blocks of HIIT can improve lower- and upper-limb aerobic and anaerobic power qualities in EJA. Tables 1, 2 and 3 provide examples of integrated maximal aerobic speed-based HIIT programs during each stage of the training plan for EJA.

Developing the Training Plan

Periodization

The deliberate manipulation of training stimuli, and the monitoring of fatigue and recovery is essential to facilitate optimal athletic performance (i.e., peaking) at pre-determined stages throughout an annual cycle (10, 13, 22, 28). Within the preparatory phases of the judo periodized program there should therefore be an initial focus on building general physical fitness, with the emphasis shifting to sport-specific conditioning closer to competition (13). The frequency of competition for judokas depends on the level and goals of the athlete. For example, during the calendar year an Indian EJA will compete in five to eight national and international competitions including state trials, national events, Asian games, European circuit events, and in some cases Olympic and World Championship events. The focus of the following sections is to provide a comprehensive training plan for EJA based on the current judo specific literature, physiological and performance requirements, and the nature of a typical annual competition schedule.

Basic Structure of the Plan

Conjugated Sequence

Conjugated sequencing, consisting of accumulation, transmutation and realization phases takes advantage of the stimulus-fatigue-recovery-adaptation principle (50). In this instance, concentrated volumes are applied for a specific period to saturate the neuromuscular system with a potent training stress, aimed at eliciting a greater super-compensation effect (22). Although the seminal Matveyev (33) model of four-week training block durations is still considered an effective paradigm, conjugated methods have shown greater efficacy in improving physiological adaptation for EJA (32). Based on this evidence, and compared to the classic periodization literature (22, 50), a depiction of two potential periodization paradigms for judo athletes is presented (Figure 1).

<Insert Figure 1>

Interestingly, Ullrich et al. (51) implemented two, four-week mesocycle blocks using a daily undulating versus conjugated model, while implementing intensities between 50-90% of 1RM to develop strength, velocity and power for sub-elite adolescent judokas. The selected exercises consisted of the squat, clean and jerk, snatch, hamstring curl, bench press, barbell bench pull, and latissimus pull-down. For both models, improvements were seen in isometric knee extensor and knee flexor maximal voluntary contraction, while upper and lower-body 1RM improvements varied from 3.5% to 13.5%. Therefore, it is likely that various periodized paradigms can improve physiological adaptations, although the conjugated sequence method may be superior, especially for elite athletic populations (13, 22).

Accumulation Phase

During the accumulation phase, the primary emphasis is to improve work capacity. As such, high volume-loads are generally prescribed to induce considerable fatigue (22). However, in judo less emphasis should be placed on muscle hypertrophy due to the weight-based nature of the sport. Instead, the impetus should be to maximize strength in order to effectively transition toward muscle power and sport-specific training stimuli within the latter phases. None the less, the volumes prescribed in this phase are still expected to induce some level of hypertrophy (41). There is a need to concurrently train the anaerobic and aerobic systems for judo athletes, and therefore HIIT sessions with variable W:R ratios at near-maximal, maximal and supramaximal intensities should be considered, similar to those recommended by Buchheitt et al (7) (Table 1).

<Insert Table 1>

Transmutation Phase

During the transmutation phase developing PPO should be prioritized. Simultaneously, a targeted and systematic reduction in training volume should allow fitness to return to baseline (23). It is recommended that maximal strength should be maintained using HST, while concurrent CL and cluster set structures are implemented to improve force-velocity characteristics. WL derivatives are also recommended during this phase to improve strength-speed, with the intention of moving moderate-to-heavy loads at high velocities to optimize RFD (12, 45, 47). As per previous phase sequencing models, athletes within the transmutation phase should be capable of producing high power outputs due to the maximal

strength developed during the accumulation phase (22, 45). Like the accumulation phase, HIIT training can be implemented concurrently to maintain aerobic and anaerobic power. However, HIIT volumes are lowered considerably here, and supramaximal intensities are prescribed with W:R ratios similar to those encountered in competition (22, 23) (Table 2).

<Insert Table 2>

Realization Phase

Within the realization phase, a ‘sport-specific’ focus should be taken, where the aim is to develop speed-strength (22). This phase is also known as the ‘taper’ or ‘specific-preparation’ phase whereby intensity is maintained and volume lowered between 41-60% of the accumulation phase, thus facilitating full recovery without compromising conditioning (22, 35). In particular, the overall aim should be to create a positive rebound in performance, whereby performance is temporarily enhanced beyond baseline. Thus, tapering, which is suggested to improve performance up to 5-6% within a 14-day ‘window’ once volumes are reduced, should be considered (35). However, there is no single superior taper strategy, and therefore the taper period should be highly individualized to the athletes’ needs. Although there is a paucity of evidence in combat sports, it appears that linear tapering (i.e., step reductions in training volumes) can markedly improve performance for taekwondo athletes (15) and thus, is likely to translate well to judo. As the ultimate goal within this phase is to optimize sports-specific RFD, the conditioning emphasis should be altered from HIIT to more sport-specific conditioning such as *Randori* (i.e., practice competition bouts) (Table 3).

<Insert Table 3>

Fatigue Monitoring and Recovery

For the strength and conditioning professional it is imperative to consider fatigue, recovery and holistic athlete wellness to facilitate effective overreaching and avoid overtraining (40). Of the currently available subjective athlete wellness questionnaires, the recovery-stress questionnaire-sport (RESTQ-S) and multi component training distress scale (MTDS) appear the most reliable and contextually valid (40). Comparatively, the MTDS has a much smaller sample of questions to the RESTQ-S (22 versus 76), which may facilitate long-term athlete adherence (39). Additionally, the MTDS is sensitive to acute alterations in workload (30), and therefore it is recommended that it be completed by athletes on a weekly basis so that

possible markers of overtraining can be detected early. In particular, it is strongly suggested that the MTDS be utilized prior to and following high volume-load work blocks, where overtraining is most likely to occur.

Practical Application

This article has demonstrated that there is a scarcity of research surrounding strength and conditioning training strategies for EJA. Based on the culmination of specific judo and other combat sport literature we have endeavoured to provide evidence-based recommendations to optimize training plans to improve EJA performance. Specifically, this article has highlighted the physiological factors governing elite judo performance, and appropriate considerations during each phase of the periodized plan. Recommendations include the implementation of partial and full ROM HST, various power training methods (i.e., CL, cluster sets, and WL derivatives) and HIIT conditioning strategies across separate appropriately sequenced phases within a competitive season. Therefore, strength and conditioning professionals should consider the sport-specific requirements and strength and conditioning recommendations presented in this paper when working with EJA.

References

1. Amtmann J and Cotton A. Strength and conditioning for judo. *Strength Cond J* 27: 26, 2005.
2. Baker D. Differences in strength and power among junior-high, senior-high, college-aged, and elite professional rugby league players. *J Strength Cond Res* 16: 581-585, 2002.
3. Baker D. Recent trends in high-intensity aerobic training for field sports. *Prof Strength Cond* 22: 3-8, 2011.
4. Baker D, Nance S, and Moore M. The load that maximizes the average mechanical power output during jump squats in power-trained athletes. *J Strength Cond Res* 15: 92-97, 2001.
5. Barbado D, Lopez-Valenciano A, Juan-Recio C, Montero-Carretero C, van Dieën JH, and Vera-Garcia FJ. Trunk stability, trunk strength and sport performance level in judo. *PloS One* 11: e0156267, 2016.
6. Bonato M, Rampichini S, Ferrara M, Benedini S, Sbriccoli P, Merati G, and Medica E. Aerobic training program for the enhancements of HR and VO₂ off-kinetics in elite judo athletes. *J Sports Med Phys Fitness* 55: 1277-1284, 2015.
7. Buchheit M and Laursen PB. High-intensity interval training, solutions to the programming puzzle. *Sports Med* 43: 313-338, 2013.

8. Campos BT, Cabido CET, Soares YM, Pedrosa GF, Mendes TT, and Almeida RSVd. A prior isometric strength training increases muscle power performance of judo athletes. *J Phys Educ* 29: 1-8, 2018.
9. Campos FAD, Bertuzzi R, Dourado AC, Santos VGF, and Franchini E. Energy demands in taekwondo athletes during combat simulation. *Eur J Appl Physiol* 112: 1221-1228, 2012.
10. Carazo-Vargas P, González-Ravé JM, Newton RU, and Moncada-Jiménez J. Periodisation model for Costa Rican taekwondo athletes. *Strength Cond J* 37: 74-83, 2015.
11. Chaabene H, Hachana Y, Franchini E, Mkaouer B, and Chamari K. Physical and physiological profile of elite karate athletes. *Sports Med* 42: 829-843, 2012.
12. Comfort P and Turner A. Developing muscular strength and power. *Advanced Strength and Conditioning*. New Dehli, Ind: Routledge, 35-60, 2017.
13. Comfort P and Turner A. Periodisation, in: *Advanced Strength and Conditioning*. New Dehli, Ind: Routledge, 116-136, 2017.
14. Davies G, Riemann BL, and Manske R. Current concepts of plyometric exercise. *Int J Sports Phys Ther* 10: 760, 2015.
15. Fortes LdS, Vasconcelos GC, Santos TMD, Paes PP, Vianna JM, and Pérez AJ. Effect of tapering on anaerobic power and capacity of tae-kwon-do athletes. *Rev bras cineantropom desempenho hum* 19: 224-232, 2017.
16. Franchini E, Artioli GG, and Brito CJ. Judo combat: time-motion analysis and physiology. *Int J Perform Anal Sport* 13: 624-641, 2013.
17. Franchini E, Del Vecchio FB, Ferreira Julio U, Matheus L, and Candau R. Specificity of performance adaptations to a periodized judo training program. *Rev Andal Med Deport* 8: 67-72, 2015.
18. Franchini E, Del Vecchio FB, Matsushigue KA, and Artioli GG. Physiological profiles of elite judo athletes. *Sports Med* 41: 147-166, 2011.
19. Franchini E, Julio UF, Panissa VL, Lira FS, Gerosa-Neto J, and Branco BH. High-intensity intermittent training positively affects aerobic and anaerobic performance in judo athletes independently of exercise mode. *Front Physiol* 7: 268, 2016.
20. Franchini E, Sterkowicz S, Szmatlan-Gabrys U, Gabrys T, and Garnys M. Energy system contributions to the special judo fitness test. *Int J Sports Physiol Perform* 6: 334-343, 2011.
21. Fukuda DH, Stout JR, Kendall KL, Smith AE, Wray ME, and Hetrick RP. The effects of tournament preparation on anthropometric and sport-specific performance measures in youth judo athletes. *J Strength Cond Res* 27: 331-339, 2013.
22. Haff GG. Roundtable discussion: Periodization of training - Part 1. *Strength Cond J* 26: 50-69, 2004.
23. Haff GG and Triplett NT. *Essentials of strength training and conditioning 4th edition*. Human Kinetics, 2015.
24. Hammami M, Negra Y, Shephard RJ, and Chelly MS. The Effect of Standard Strength vs. Contrast Strength Training on the Development of Sprint, Agility,

- Repeated Change of Direction, and Jump in Junior Male Soccer Players. *J Strength Cond Res* 31: 901-912, 2017.
25. Helm N, Prieske O, Muehlbauer T, Krüger T, and Granacher U. Effects of judo-specific resistance training on kinetic and electromyographic parameters of pulling exercises in judo athletes. *Sportverletz Sportschaden*, 32: 134-142, 2018.
26. Herman SL and Smith DT. Four-Week Dynamic Stretching Warm-up Intervention Elicits Longer-Term Performance Benefits. *J Strength Cond Res* 22: 1286-1297, 2008.
27. Iglesias-Soler E, Carballeira E, Sánchez-Otero T, Mayo X, and Fernández-del-Olmo M. Performance of maximum number of repetitions with cluster-set configuration. *Int J Sports Physiol Perform* 9: 637-642, 2014.
28. James LP, Kelly VG, and Beckman EM. Periodisation for Mixed Martial Arts. *Strength Cond J* 35: 34-45, 2013.
29. Maffiuletti NA, Aagaard P, Blazevich AJ, Folland J, Tillin N, and Duchateau J. Rate of force development: physiological and methodological considerations. *Eur J Appl Physiol* 116: 1091-1116, 2016.
30. Main L and Grove JR. A multi-component assessment model for monitoring training distress among athletes. *Eur J Sport Sci* 9: 195-202, 2009.
31. Malliaropoulos NG, Callan M, and Johnson J. Comprehensive training programme for judo players nine plus 9+: possible lower limb primary injury prevention. *Muscles Ligaments Tendons J* 4: 262, 2014.
32. Marques L, Franchini E, Drago G, Aoki M and Moreira A. Physiological and performance changes in national and international judo athletes during block periodization training. *Biol Sport* 34: 371–378, 2017.
33. Matveyev L. Periodizing Sport Training. Berlin: Berles & Wertniz, 1972.
34. Monteiro L. Analysis of rate force development, power and resistance explosive strength indicators in top elite vs. elite male judokas. *App Res Judo* 36 2016.
35. Mujika I, Berryman N, and Bosquet L. Managing the training load of overreached athletes: Insights from the detraining and tapering literature. *Sport, Recovery, and Performance*. New Dehli, Ind: Routledge, 101-121, 2017.
36. Pocecco E, Ruedl G, Stankovic N, Sterkowicz S, Del Vecchio FB, Gutiérrez-García C, Rousseau R, Wolf M, Kopp M, and Miarka B. Injuries in judo: a systematic literature review including suggestions for prevention. *Br J Sports Med* 47: 1139-1143, 2013.
37. Ramirez-Campillo R, Álvarez C, García-Hermoso A, Ramírez-Vélez R, Gentil P, Asadi A, Chaabene H, Moran J, Meylan C, and García-de-Alcaraz A. Methodological Characteristics and Future Directions for Plyometric Jump Training Research: A Scoping Review. *Sports Med* 48:1059-1081, 2018.
38. Rhea MR, Kenn JG, Peterson MD, Massey D, Simão R, Marin PJ, Favero M, Cardozo D, and Krein D. Joint-angle specific strength adaptations influence improvements in power in highly trained athletes. *Hum Mov* 17: 43-49, 2016.
39. Saw AE, Main LC, and Gustin PB. Monitoring athletes through self-report: factors influencing implementation. *J Sports Sci Med* 14: 137, 2015.

40. Saw AE, Main LC, and Gastin PB. Monitoring the athlete training response: subjective self-reported measures trump commonly used objective measures: a systematic review. *Br J Sports Med* 50: 281-291, 2016.
41. Schoenfeld BJ, Grgic J, Ogborn D, and Krieger JW. Strength and hypertrophy adaptations between low-vs. high-load resistance training: a systematic review and meta-analysis. *J Strength Cond Res* 31: 3508-3523, 2017.
42. Shield AJ and Bourne MN. Hamstring Injury Prevention Practices in Elite Sport: Evidence for Eccentric Strength vs. Lumbo-Pelvic Training. *Sports Med* 48: 513-524, 2018.
43. Sikorski W, Mickiewicz G, Majle B, Laksa C. Structure of the contest and work capacity of the judoist. Proceedings of the International Congress on Judo: Contemporary Problems of Training and Judo Contest. Warsaw, Spala-Poland: 9-11, 1987.
44. Stone MH, O'bryant HS, Mccoy L, Coglianese R, Lehmkuhl M, and Schilling B. Power and maximum strength relationships during performance of dynamic and static weighted jumps. *J Strength Cond Res* 17: 140-147, 2003.
45. Suchomel TJ, Comfort P, and Lake JP. Enhancing the force-velocity profile of athletes using weightlifting derivatives. *Strength Cond J* 39: 10-20, 2017.
46. Suchomel TJ, Nimphius S, Bellon CR, and Stone MH. The Importance of Muscular Strength: Training Considerations. *Sports Med* 48: 765-785, 2018.
47. Suchomel TJ and Sole CJ. Force-Time-Curve Comparison Between Weight-Lifting Derivatives. *Int J Sports Physiol Perform* 12: 431-439, 2017.
48. Takahashi R. Power training for judo: plyometric training with medicine balls. *Strength Cond J* 14: 66-71, 1992.
49. Tufano JJ, Brown LE, and Haff GG. Theoretical and practical aspects of different cluster set structures: a systematic review. *J Strength Cond Res* 31: 848-867, 2017.
50. Turner A. The science and practice of periodization: a brief review. *Strength Cond J* 33: 34-46, 2011.
51. Ullrich B, Pelzer T, Oliveira S, Pfeiffer M. Neuromuscular responses to short-term resistance training with traditional and daily undulating periodization in adolescent elite judoka. *J Strength Cond Res* 30: 2083-2099, 2016.
52. Wilson ME. The daily adjustable progressive resistance exercise system: Getting reacquainted with an old friend. *Strength Cond J* 30: 76-78, 2008.

Figure 1. Depicts an example of (A) a traditional periodisation paradigm compared to (B) a conjugated sequence periodisation paradigm. Volume-load is expressed as the sum of sets x repetitions x weight lifted, where weight lifted is prescribed as a percentage of 1-RM.

Table 1. Resistance training and aerobic conditioning prescription for the accumulation phase within a competitive year for elite Judo athletes (EJA). Tu: Tuesday, Sa: Saturday, Th: Thursday, 1RM: one-repetition maximum, MAS: maximal aerobic speed.

Table 2. Resistance training and aerobic conditioning prescription for the transmutation phase within a competitive year for elite Judo athletes (EJA). Tu: Tuesday, Sa: Saturday, Th: Thursday, 1RM: one-repetition maximum, MAS: maximal aerobic speed.

Table 3. Resistance training and aerobic conditioning prescription for the realisation phase within a competitive year for elite Judo athletes (EJA). Tu: Tuesday, Sa: Saturday, Th: Thursday, 1RM: one-repetition maximum, MAS: maximal aerobic speed.

Resistance Training Prescription for Accumulation Phase					
Tuesday/Saturday		Thursday			
Power Clean ^a Box Squat Deadlift Split Squat SL Romanian Deadlift		Push Jerk ^a Prone Bench Row Flat Bench Press Single Arm Bent-Over Row ^b Landmine Push Press Pull Up			
Loading Scheme					
Week of Block	Sets	Reps	% of 1RM		
1	4	10	Tu/Sa: 70%, Th: 65%		
2	4	6	Tu/Sa: 80%, Th: 85%		
3	4	10	Tu/Sa: 70%, Th: 65%		
4	4	6	Tu/Sa: 80%, Th: 85%		
^a Power maintenance: 4 sets x 6 reps at 50% of 1RM					
^b Strength endurance maintenance: 5 sets x 12reps at 65% of 1RM					
Aerobic Conditioning Prescription for Accumulation Phase					
Mode of Exercise	Set Length	Work Intervals	Recovery	Work:Rest	Number of Repetitions
Rowing	5 mins	30 s @ 100% MAS	30 s @ 60-70% MAS	1:1 Active recovery	5
Cycling	5 mins	30 s @ 100% MAS	30 s @ 60-70% MAS	1:1 Active recovery	5
Arm Ergometer	5 mins	30 s @ 100% MAS	30 s @ 60-70% MAS	1:1 Active recovery	5

Resistance Training Prescription for Transmutation Phase					
Tuesday/Saturday		Thursday			
Hang Power Clean and Push Press Squat ^a Box Jumps (60 cm) ^c Incline Bench Press Medicine Ball Squat and Throw		Power Clean ^b Mid-thigh Snatch Hex Bar Squats ^d Depth Jumps ^c			
Contrast Loading Method		Contrast Loading Method			
Loading Scheme					
Week of Block	Sets	Reps	% of 1RM		
1	4	8	Tu/Sa: 80%, Th: 80%		
2	3	6	Tu/Sa: 85%, Th: 85%		
3	4	8	Tu/Sa: 80%, Th: 80%		
^a Maximal strength maintenance: 3 sets x 5 reps at 85% of 1RM ^b Power Clean: 3 sets x 5 reps at 30% 1RM ^c Plyometrics: body weight for 20 foot contacts on Tu/Th/Sa x 5 sets ^d Hex Bar Squats using clusters: 3 sets x 6 reps at 85% of 1RM, using 3 rep clusters with 10-15sec intra-set rest					
Aerobic Conditioning Prescription for Transmutation Phase					
Mode of Exercise	Set Length	Work Intervals	Recovery	Work:Rest	Number of Repetitions
Rowing	4 mins	30s @ 105- 110% MAS	10s passive recovery	3:1 (passive recovery)	6
Cycling	4 mins	30s @ 105- 110% MAS	10s passive recovery	3:1 (passive recovery)	6
Arm Ergometer	4 mins	30s @ 105- 110% MAS	10s passive recovery	3:1 (passive recovery)	6

Resistance Training Prescription for Realisation Phase					
Tuesday/Saturday		Thursday			
Mid-thigh Snatch Hip Thruster ^a Bridge and Toss ^d (dead ball) (<i>osaekomi waza</i>) Bench Throws (smith machine) ^c		Jump Shrug ^b Hang High Pull ^b Deadlift ^a Clean Pull From Floor			
Loading Scheme					
Week of Block	Sets	Reps	% of 1RM		
1	3	6	T/S: 80%, Th: 80%		
2	3	6	T/S: 80%, Th: 80%		
^a Maximal strength maintenance: 3 sets x 5 reps at 85% of 1RM ^b Velocity: 3 sets x 6 reps at 30% of 1RM, to develop the <i>velocity end</i> of the force-velocity curve ^c Bench Throws: 3 sets x 6 reps at ~28% of 1RM ^d Dead-Ball: 3 sets x 10reps with 20kg Dead-ball					
Aerobic Conditioning Prescription for Realisation Phase					
Mode of Exercise	Set Length	Work Intervals	Recovery	Work:Rest	Number of Repetitions/Bouts
Rowing	4 min	30 s @ 105- 110% MAS	10 s Passive Rest	3:1 (passive recovery)	6
<i>Randori</i>	9 min	Full contact sparring (3 mins bouts)	No rest during the bout – 3 min between bouts	1:1	3

